📘 SQL Set Operators

🔹 Rules for Set Operators

Rule 1: Usage in Clauses

- Set operators can be used with most SQL clauses:
- `WHERE`
- `JOIN`
- `GROUP BY`
- `HAVING`
- ❗ `ORDER BY` can be used **only once**, at the **end of the final query**

Example:

```
SELECT
    FIRSTNAME,
    LASTNAME
FROM CUSTOMERS
JOIN ...
WHERE ...
GROUP BY ...

UNION

SELECT
    FIRSTNAME,
    LASTNAME
FROM CUSTOMERS
JOIN ...
WHERE ...
GROUP BY ...

ORDER BY FIRSTNAME;  -- Only allowed at the end
```

Rule 2: Same Number of Columns

- Each query must return the **same number of columns**

```
SELECT col1, col2 FROM table1
UNION
SELECT col1 FROM table2;  -- ❌ ERROR
```

Rule 3: Compatible Data Types

- Corresponding columns must have **compatible data types**
- INT ↔ INT
- VARCHAR ↔ VARCHAR

Rule 4: Same Column Order

- Column order must match across queries

```
SELECT name, age FROM table1
UNION
SELECT age, name FROM table2;  -- ❌ Incorrect mapping
```

Rule 5: Column Names

- Final result column names are taken from the **first query**

🔹 Types of Set Operators

1. UNION

- Combines results from multiple queries
- Removes duplicates

```
SELECT CITY FROM STATION1
UNION
SELECT CITY FROM STATION2;
```

✔ Returns unique cities from both tables

---

2. UNION ALL

- Combines results
- Keeps duplicates

```
SELECT CITY FROM STATION1
UNION ALL
SELECT CITY FROM STATION2;
```

✔ Returns all rows including duplicates

---

3. EXCEPT

- Returns rows present in the first query but not in the second

```
SELECT CITY FROM STATION1
EXCEPT
SELECT CITY FROM STATION2;
```

✔ Cities in `STATION1` but not in `STATION2`

Alternative:

```
SELECT CITY
FROM STATION1
WHERE CITY NOT IN (SELECT CITY FROM STATION2);
```

---

4. INTERSECT

- Returns common rows between queries

```
SELECT CITY FROM STATION1
INTERSECT
SELECT CITY FROM STATION2;
```

✔ Cities present in both tables

❗ **MySQL Note:**

- MySQL does **NOT** support `INTERSECT` directly

Alternative:

```
SELECT CITY
FROM STATION1
WHERE CITY IN (SELECT CITY FROM STATION2);
```

🔹 Joins vs Set Operators

Concept	Joins	Set Operators
Purpose	Combine columns	Combine rows
Based on	Relationship (keys)	Independent queries
Output Shape	Wider table	Same structure table

Concept	Joins	Set Operators
Common Usage	Relational data	Comparing datasets

🔹 When to Use

✅ Use JOINS when:

- Tables are **related**
- You want to **combine columns**
- There is a **common key**

✅ Use Set Operators when:

- Tables have the **same structure**
- You want to **combine or compare datasets**
- No relationship is required

🔹 Quick Use Cases

Use Case	Recommended Approach
Combine related tables	JOIN
Merge similar datasets	UNION
Find common rows	INTERSECT
Find differences	EXCEPT
